

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

**An Efficient Deep Learning Approach for Bean Leaf Disease Recognition with
ResNet152V2**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**An Efficient Deep Learning Approach for Bean Leaf Disease Recognition with ResNet152V2**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Monika . The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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Date: 15-05-2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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1. Introduction

Agriculture plays a vital role in food production and the economy. Bean crops are widely cultivated and consumed across the world. However, bean plants are affected by several diseases that reduce crop quality and yield. Some common bean leaf diseases include Angular Leaf Spot and Bean Rust.

Traditional disease detection methods depend on manual inspection by experts, which is time-consuming and sometimes inaccurate. Early disease identification is important to prevent crop loss and improve agricultural productivity.

With the advancement of Artificial Intelligence and Deep Learning, automated disease detection systems can be developed for faster and more accurate diagnosis. In this project, we use the ResNet152V2 deep learning model for bean leaf disease classification.

1.1 Objectives

- Detect bean leaf diseases automatically using AI
- Classify images into Angular Leaf Spot, Bean Rust, and Healthy classes
- Improve disease detection accuracy using ResNet152V2
- Reduce crop loss through early diagnosis
- Support smart and sustainable agriculture

1.2 Problem Statement

Bean leaf diseases reduce crop productivity and quality. Manual disease detection requires expert knowledge and large-scale monitoring is difficult. Delayed detection increases crop damage and financial loss.

Therefore, there is a need for an automated AI-based disease detection system that provides fast and accurate classification.

2. METHODOLOGY

The proposed system follows a deep learning-based methodology for bean leaf disease classification.

Steps Involved

1. Dataset Collection
2. Image Preprocessing
3. Data Augmentation
4. Model Training using ResNet152V2
5. Validation and Testing
6. Performance Evaluation

The dataset images were resized and preprocessed before training. Data augmentation techniques such as rotation, zooming, and flipping were applied to improve model performance.

3. TOOLS AND TECHNOLOGIES

The following tools and technologies were used in this project:

Tool / Technology	Purpose
Python	Programming Language
TensorFlow / Keras	Deep Learning Framework
ResNet152V2	CNN Model
OpenCV	Image Processing
NumPy	Numerical Operations
Matplotlib	Data Visualization

Flask

Web Application Framework

VS Code

Development Environment

4. IMPLEMENTATION

The implementation phase involved training and testing the deep learning model using bean leaf images.

Dataset

The dataset was collected from Kaggle and consisted of three classes:

- Angular Leaf Spot
- Bean Rust
- Healthy

Dataset Distribution

- Training Images: 1034
- Validation Images: 133

Model Used

ResNet152V2 is a deep Convolutional Neural Network containing 152 layers. It uses residual connections to solve the vanishing gradient problem and improve learning efficiency.

Features of ResNet152V2

- Deep CNN architecture
- High feature extraction capability
- Better classification performance
- Efficient training through residual learning

The model analyzes bean leaf images and predicts the disease category.

5. MAJOR FINDINGS / RESULTS

The ResNet152V2 model achieved strong classification performance on the validation dataset.

Performance Metrics

Metric	Result
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Accuracy	88%
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Precision	0.85 – 0.92
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Recall	0.84 – 0.91
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F1-Score	0.85 – 0.90
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The confusion matrix showed minimal misclassification, indicating that the model

generalized well on unseen data.

Advantages of the Proposed System

- Fast disease detection
- Reduced manual effort
- Early crop monitoring
- Better agricultural productivity
- Supports smart farming applications

6. CONCLUSION AND FUTURE SCOPE

This project successfully developed a deep learning-based bean leaf disease recognition system using the ResNet152V2 model.

The model accurately classified bean leaf diseases into Angular Leaf Spot, Bean Rust, and Healthy categories with an overall validation accuracy of 88%.

The project demonstrates how Artificial Intelligence can improve agriculture through automatic disease detection and crop monitoring.

Future Scope

- Increase dataset size for better accuracy
- Develop real-time mobile applications
- Integrate IoT-based smart farming systems

- Compare with advanced deep learning architectures

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APPENDICES

Appendix A – Screenshots

- Training Accuracy Graph
- Validation Accuracy Graph
- Confusion Matrix
- Web Application Interface

Appendix B – Source Code

- app.py
- bean_leaf_model.py
- HTML templates

